Lecture 3 - Learning Basics

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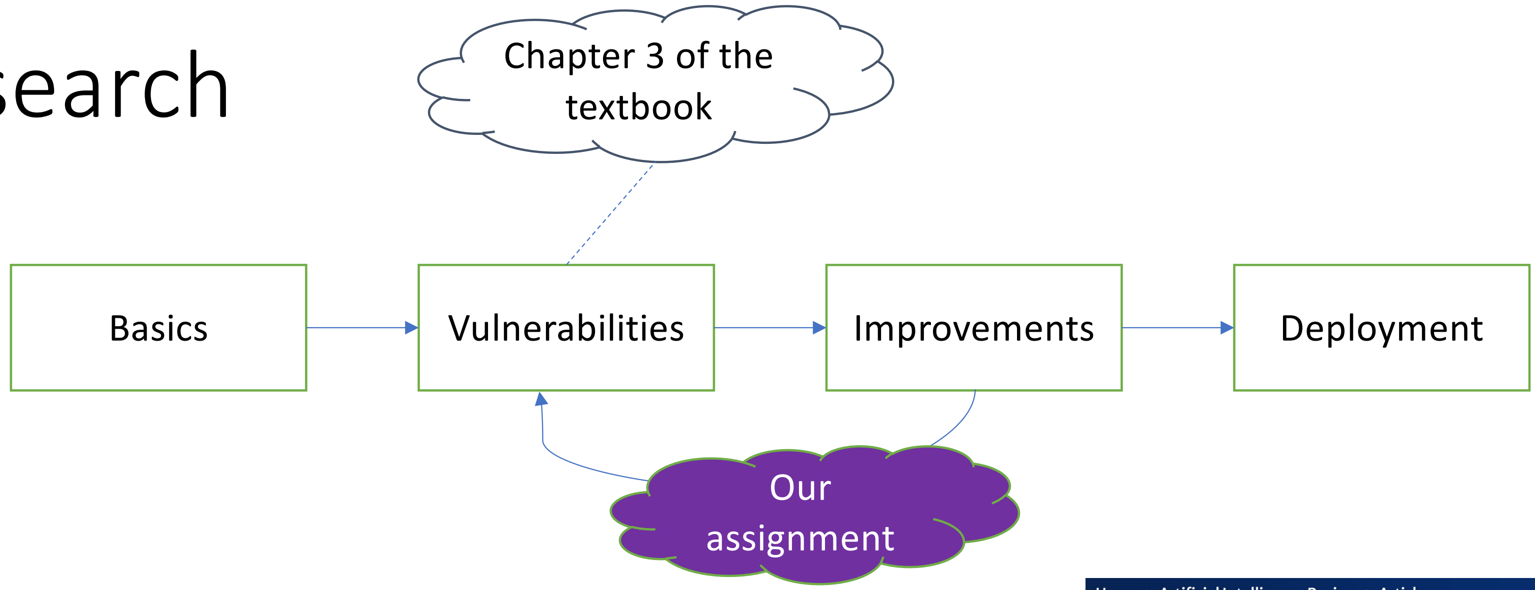
(Attendance Code: **665240**)

Research



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Survey

A survey of safety and trustworthiness of deep neural networks: Verification, testing, adversarial attack and defence, and interpretability ☆

Xiaowei Huang [a], Daniel Kroening [b], Wenjie Ruan [c], James Sharp [d], Youcheng Sun [e], Emese Thamo [a], Min Wu [b], Xinping Yi [a]

Entering LLM/GenAI

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A survey of safety and trustworthiness of large language models through the lens of verification and validation

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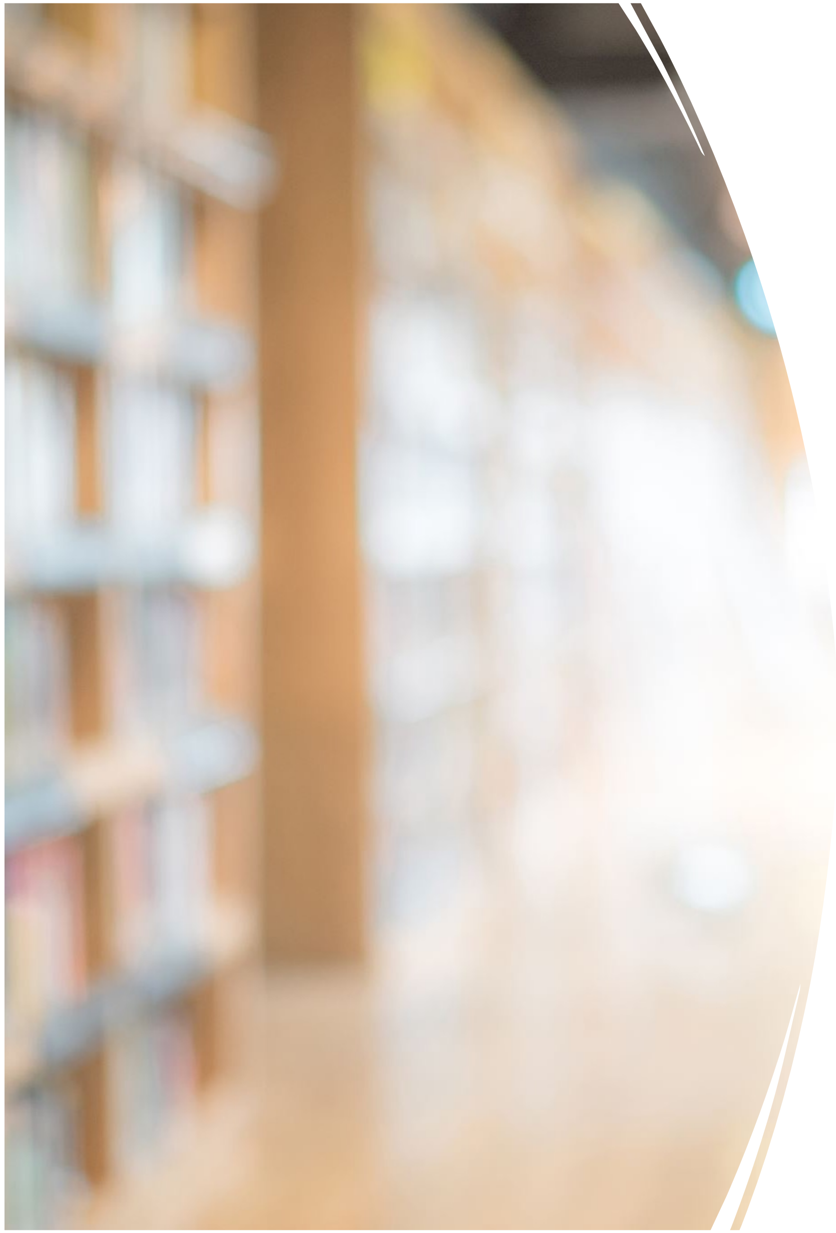
Xiaowei Huang, Wenjie Ruan, Wei Huang, Gaojie Jin, Yi Dong, Changshun Wu, Saddek Bensalem, Ronghui Mu, Yi Qi, Xingyu Zhao, Kaiwen Cai, Yanghao Zhang, Sihao Wu, Peipei Xu, Dengyu Wu, Andre Freitas & Mustafa A. Mustafa

In the last lecture,

- What is machine learning?
- A few applications of machine learning
- consider how to represent instances as fixed-length feature vectors



Topics

- Learning basics:
  - Before learning: data collection
  - Learning tasks: supervised and unsupervised learning
  - Learning schemes

Before learning: data collection

Independent and identically distributed (i.i.d.)

- we often assume that training instances are *independent and identically distributed* (i.i.d.) – sampled independently from the same unknown distribution
- A set of random variables $X_1, X_2, \ldots, X_n$ are i.i.d. if:
  - They are **independent** (no influence on each other).
  - They are **identically distributed** (drawn from the same distribution).

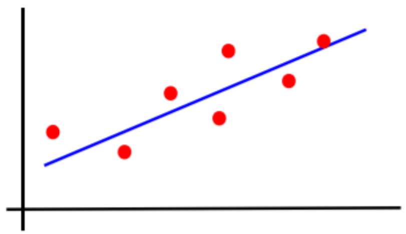
Independent and identically distributed (i.i.d.)

- we often assume that training instances are *independent and identically distributed* (i.i.d.) – sampled independently from the same unknown distribution
- there are also cases where this assumption does not hold
  - cases where sets of instances have dependencies
    - instances sampled from the same medical image
    - instances from time series
    - etc.

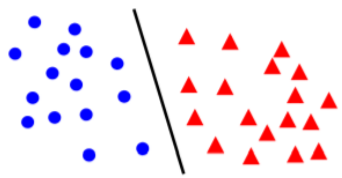
Learning tasks: supervised and unsupervised learning

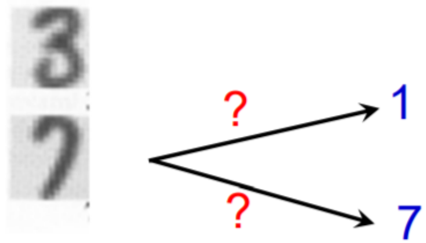
Three canonical learning problems

1. Regression - supervised
   - estimate parameters, e.g. of weight vs height



2. Classification - supervised
   - estimate class, e.g. handwritten digit classification

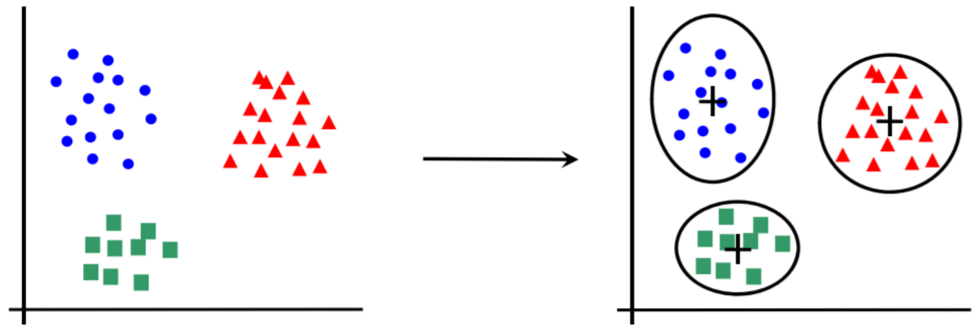




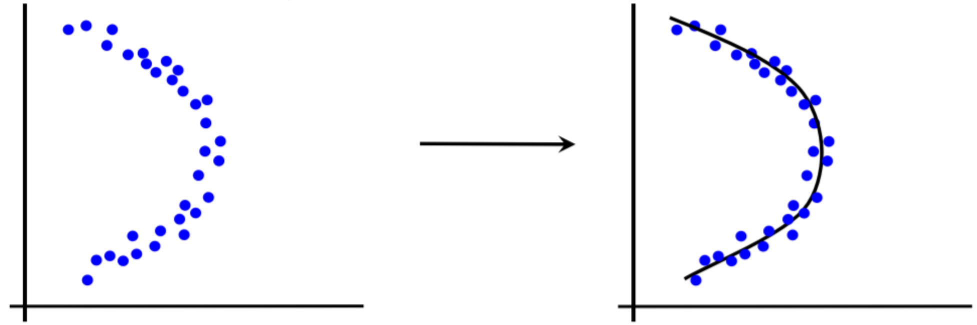
Three canonical learning problems

3. Unsupervised learning – model the data

- clustering



- dimensionality reduction



The supervised learning task

- problem setting
  - set of possible instances: *X*
  - unknown *target function:* $f : X \rightarrow Y$
  - set of *models* (a.k.a. *hypotheses*): $H = \{h \mid h : X \rightarrow Y\}$

$f(x) = w_1x+w_2$

$f(x) = (w_1x+w_2)>10$

- given *training set* of instances of unknown target function f

$$\left(\mathbf{x}^{(1)}, y^{(1)}\right), \left(\mathbf{x}^{(2)}, y^{(2)}\right) \ldots \left(\mathbf{x}^{(m)}, y^{(m)}\right)$$

- Output
  - model $h \in H$ that best approximates target function

The supervised learning task

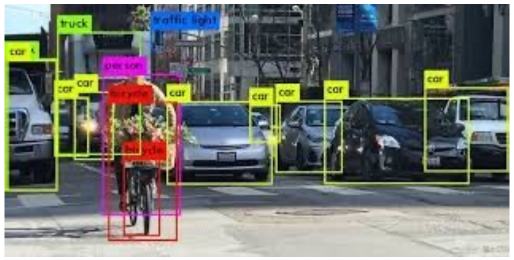
$f(x) = (w_1x+w_2)>10$

- when y is discrete, we term this a *classification* task (or *concept learning*)

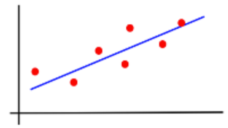
- when y is continuous, it is a *regression* task

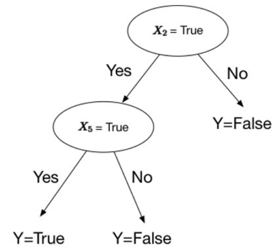
$f(x) = w_1x+w_2$

- there are also tasks in which each y is more structured object like a *sequence* of discrete labels (as in e.g. image segmentation, machine translation)

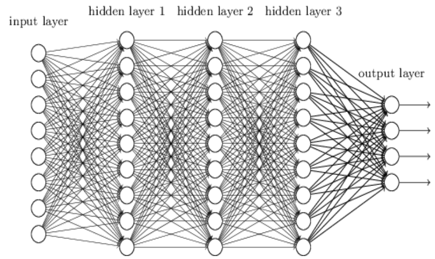


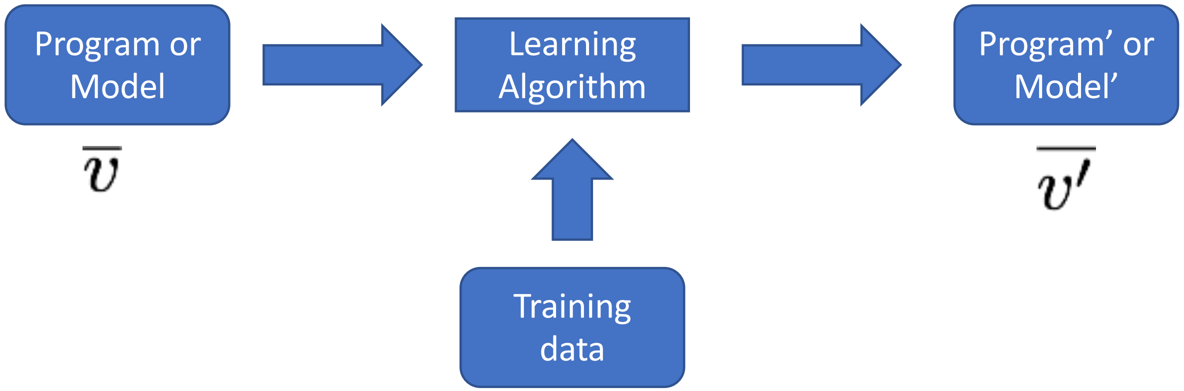
Model representations





- throughout the semester, we will consider a broad range of representations for learned models (i.e., hypothesis space), including
  - decision trees
  - neural networks
  - Linear/logistic regression
  - Bayesian networks
  - etc.





Mushroom features (from the UCI Machine Learning Repository)

Input features

Values of the features

sunken is one possible value of the *cap-shape* feature

cap-shape: bell=b,conical=c,convex=x,flat=f, knobbed=k,sunken=s
cap-surface: fibrous=f,grooves=g,scaly=y,smooth=s
cap-color: brown=n,buff=b,cinnamon=c,gray=g,green=r, pink=p,purple=u,red=e,white=w,yellow=y
bruises?: bruises=t,no=f
odor: almond=a,anise=l,creosote=c,fishy=y,foul=f, musty=m,none=n,pungent=p,spicy=s
gill-attachment: attached=a,descending=d,free=f,notched=n
gill-spacing: close=c,crowded=w,distant=d
gill-size: broad=b,narrow=n
gill-color: black=k,brown=n,buff=b,chocolate=h,gray=g, green=r,orange=o,pink=p,purple=u,red=e, white=w,yellow=y
stalk-shape: enlarging=e,tapering=t
stalk-root: bulbous=b,club=c,cup=u,equal=e, rhizomorphs=z,rooted=r,missing=?
stalk-surface-above-ring: fibrous=f,scaly=y,silky=k,smooth=s
stalk-surface-below-ring: fibrous=f,scaly=y,silky=k,smooth=s
stalk-color-above-ring: brown=n,buff=b,cinnamon=c,gray=g,orange=o, pink=p,red=e,white=w,yellow=y
stalk-color-below-ring: brown=n,buff=b,cinnamon=c,gray=g,orange=o, pink=p,red=e,white=w,yellow=y
veil-type: partial=p,universal=u
veil-color: brown=n,orange=o,white=w,yellow=y
ring-number: none=n,one=o,two=t
ring-type: cobwebby=c,evanescent=e,flaring=f,large=l, none=n,pendant=p,sheathing=s,zone=z
spore-print-color: black=k,brown=n,buff=b,chocolate=h,green=r, orange=o,purple=u,white=w,yellow=y
population: abundant=a,clustered=c,numerous=n, scattered=s,several=v,solitary=y
habitat: grasses=g,leaves=l,meadows=m,paths=p, urban=u,waste=w,woods=d

A learned decision tree

Root $\rightarrow$ leaf

```
odor = a: e (400.0)
odor = c: p (192.0)
odor = f: p (2160.0)
odor = l: e (400.0)
odor = m: p (36.0)
odor = n
|    spore-print-color = b: e (48.0)
|    spore-print-color = h: e (48.0)
|    spore-print-color = k: e (1296.0)
|    spore-print-color = n: e (1344.0)
|    spore-print-color = o: e (48.0)
|    spore-print-color = r: p (72.0)
|    spore-print-color = u: e (0.0)
|    spore-print-color = w
|    |    gill-size = b: e (528.0)
|    |    gill-size = n
|    |    |    gill-spacing = c: p (32.0)
|    |    |    gill-spacing = d: e (0.0)
|    |    |    gill-spacing = w
|    |    |    |    population = a: e (0.0)
|    |    |    |    population = c: p (16.0)
|    |    |    |    population = n: e (0.0)
|    |    |    |    population = s: e (0.0)
|    |    |    |    population = v: e (48.0)
|    |    |    |    population = y: e (0.0)
|    spore-print-color = y: e (48.0)
odor = p: p (256.0)
odor = s: p (576.0)
odor = y: p (576.0)
```

if odor=almond, predict edible

if odor=none $\land$
spore-print-color=white $\land$
gill-size=narrow $\land$
gill-spacing=crowded,
predict poisonous

Classification with a learned decision tree

x = <bell,fibrous,brown,false, foul,...>

```
odor = a: e (400.0)
odor = c: p (192.0)
odor = f: p (2160.0)
odor = l: e (400.0)
odor = m: p (36.0)
odor = n
|   spore-print-color = b: e (48.0)
|   spore-print-color = h: e (48.0)
|   spore-print-color = k: e (1296.0)
|   spore-print-color = n: e (1344.0)
|   spore-print-color = o: e (48.0)
|   spore-print-color = r: p (72.0)
|   spore-print-color = u: e (0.0)
|   spore-print-color = w
|   |   gill-size = b: e (528.0)
|   |   gill-size = n
|   |   |   gill-spacing = c: p (32.0)
|   |   |   gill-spacing = d: e (0.0)
|   |   |   gill-spacing = w
|   |   |   |   population = a: e (0.0)
|   |   |   |   population = c: p (16.0)
|   |   |   |   population = n: e (0.0)
|   |   |   |   population = s: e (0.0)
|   |   |   |   population = v: e (48.0)
|   |   |   |   population = y: e (0.0)
|   spore-print-color = y: e (48.0)
odor = p: p (256.0)
odor = s: p (576.0)
odor = y: p (576.0)
```

Trained decision tree

y = ?

y=p

Unsupervised learning

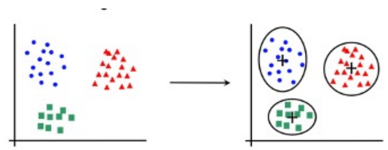
- in unsupervised learning, we're given a set of instances, without y's

$$\mathbf{x}^{(1)}, \mathbf{x}^{(2)} \dots \mathbf{x}^{(m)}$$

goal: discover interesting regularities/structures/patterns that characterize the instances

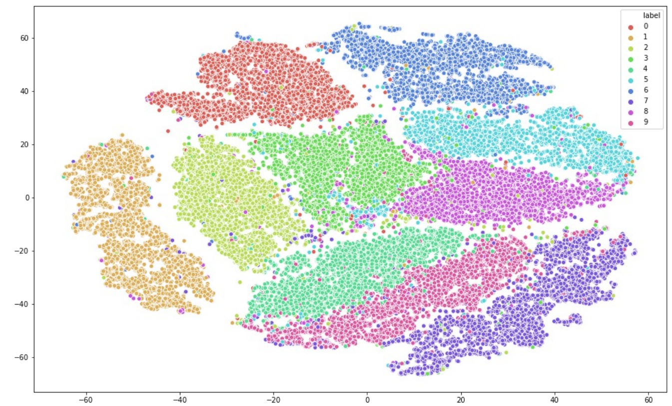
- common unsupervised learning tasks
  - *clustering*
  - *anomaly detection*
  - *dimensionality reduction*

Clustering

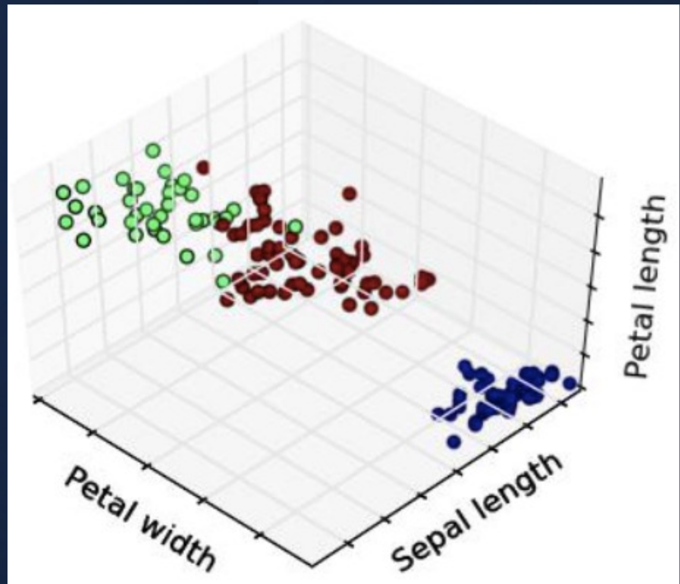


- given
  - training set of instances $\mathbf{x}^{(1)}$, $\mathbf{x}^{(2)}$ ... $\mathbf{x}^{(m)}$

- output
  - model $h \in H$ that divides the training set into clusters such that there is intra-cluster similarity and inter-cluster dissimilarity

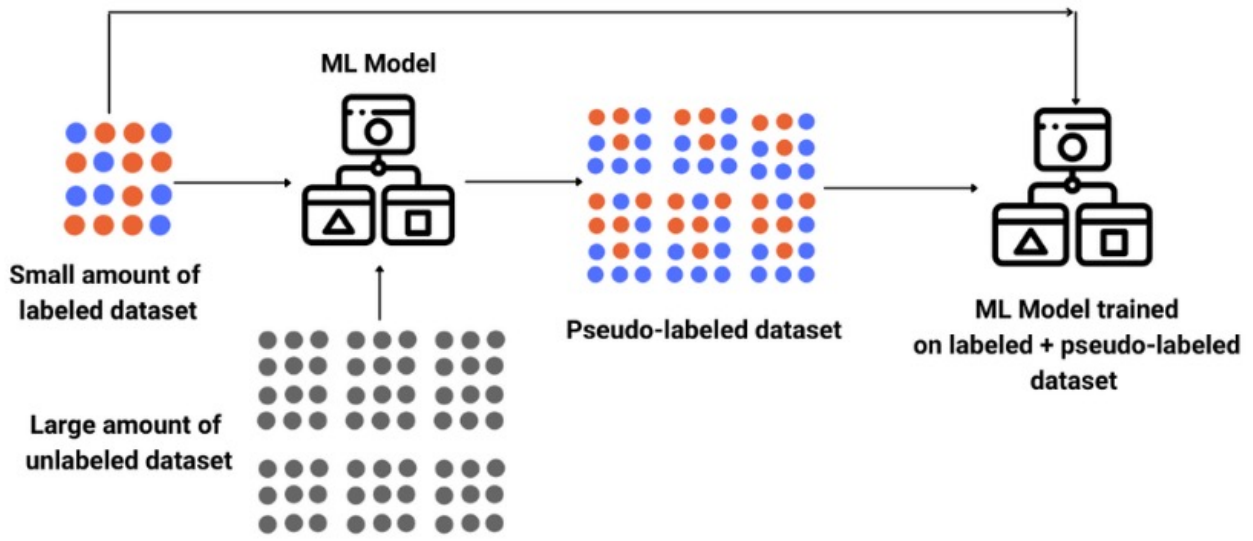


Clustering example



Semi-Supervised Learning

- using labelled as well as unlabelled data to perform certain learning tasks



Anomaly detection

performance task

given

- a previously unseen $\boldsymbol{x}$

determine

- if $\boldsymbol{x}$ looks normal or anomalous

learning task

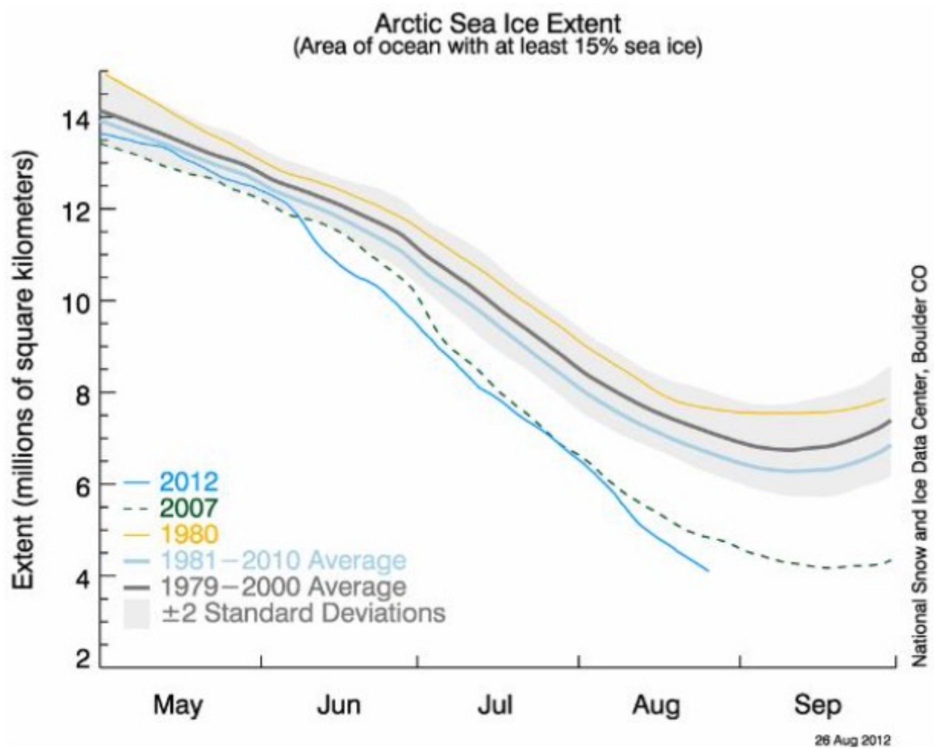
given

- training set of instances $\mathbf{x}^{(1)}, \mathbf{x}^{(2)} \ldots \mathbf{x}^{(m)}$

output

- model $h \in H$ that represents "normal" $\boldsymbol{x}$

Anomaly detection example

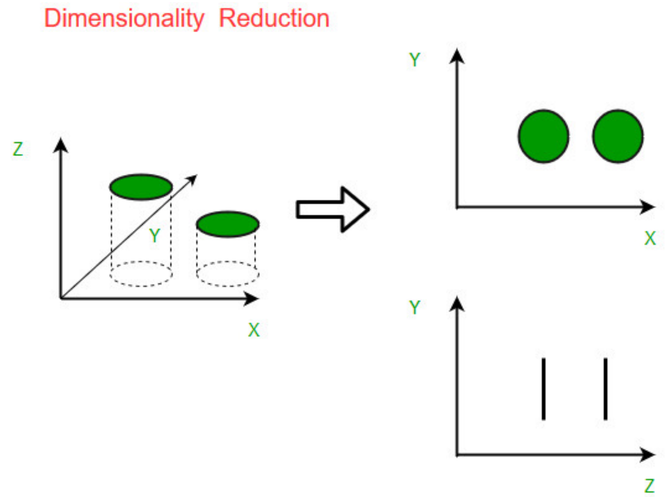


Let's say our model is represented by: 1979-2000 average, ±2 stddev.

Does the data for 2012 look anomalous?

Dimensionality reduction

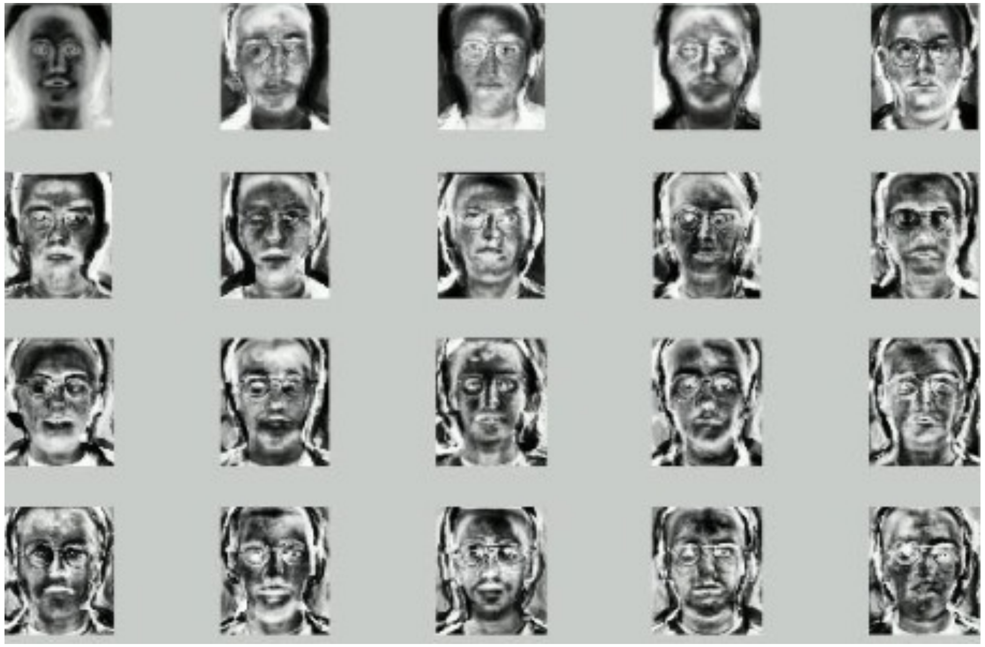
- given
  - training set of instances $\mathbf{x}^{(1)}$, $\mathbf{x}^{(2)}$ ... $\mathbf{x}^{(m)}$



- output
  - Model $h \in H$ that represents each x with a lower-dimension feature vector while still preserving key properties of the data

Dimensionality reduction example

We can represent a face using all of the pixels in a given image



More effective method (for many tasks): represent each face as a linear combination of *eigenfaces*

Dimensionality reduction example

- represent each face as a linear combination of *eigenfaces*

 $= \alpha_1^{(1)} \times$  $+ \alpha_2^{(1)} \times$  $+ \ldots + \alpha_{20}^{(1)} \times$ 

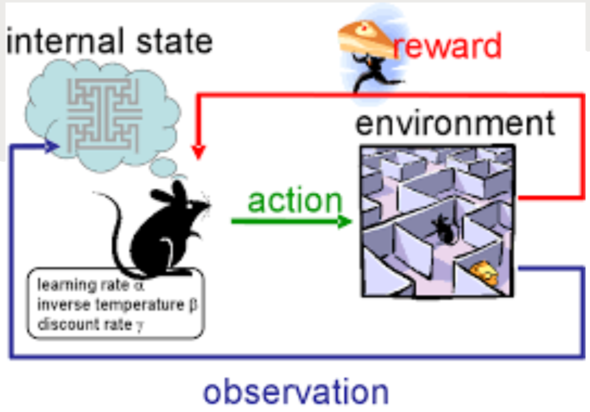
$$\mathbf{x}^{(1)} = \left\langle \alpha_1^{(1)}, \alpha_2^{(1)}, \ldots, \alpha_{20}^{(1)} \right\rangle$$

 $= a_1^{(2)}$ ´  $+ a_2^{(2)}$ ´  $+ \ldots + \alpha_{20}^{(2)} \times$ 

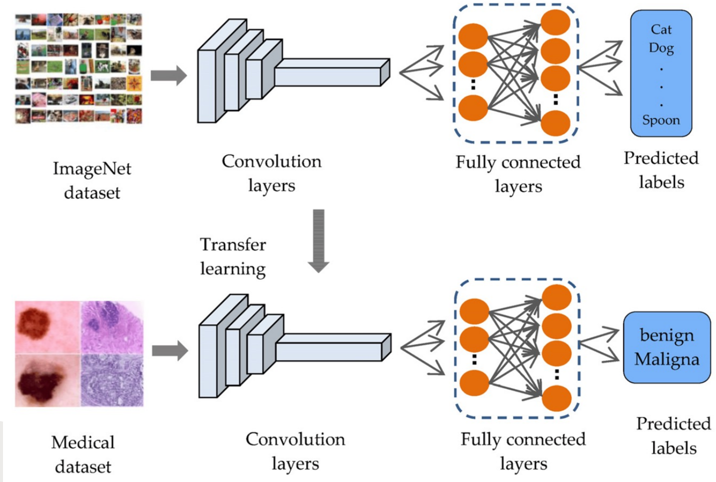
$$\mathbf{x}^{(2)} = \left\langle \alpha_1^{(2)}, \alpha_2^{(2)}, \ldots, \alpha_{20}^{(2)} \right\rangle$$

- # of features is now 20 instead of # of pixels in images

Other learning tasks



- later in the semester we'll cover other learning tasks that are not strictly supervised or unsupervised
  - *reinforcement learning*
  - *transfer learning*
  - *etc.*



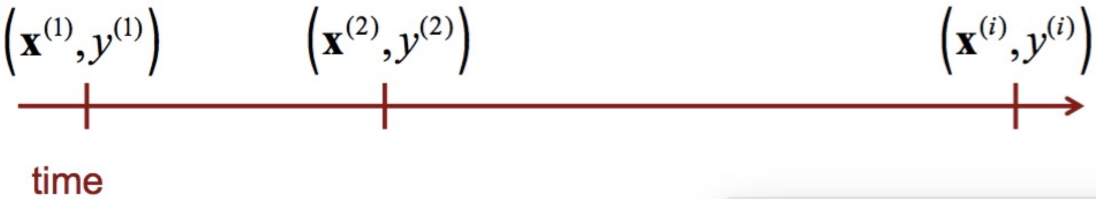
Training Schemes

Batch vs. online learning

- In batch learning, the learner is given the training set as a batch (i.e. all at once)

$$\left(\mathbf{x}^{(1)}, y^{(1)}\right), \left(\mathbf{x}^{(2)}, y^{(2)}\right) \ldots \left(\mathbf{x}^{(m)}, y^{(m)}\right)$$

- In online learning, the learner receives instances sequentially, and updates the model after each (for some tasks it might have to classify/make a prediction for each x(i) before seeing y(i))



Active learning and concept drift

Active learning: cases where the learner can select which instances for training

the target function changes over time (*concept drift*)

Learning from feedback

- RLHF (Reinforcement Learning from **Human** Feedback) trains AI by using direct human preferences to guide the model,

How to scale it?

- RLAIF (Reinforcement Learning from **AI** Feedback) uses an AI model to generate preference labels instead of humans

How to ensure alignment?

Generalization

- The primary objective in supervised learning is to find a model that *generalizes*
  - one that accurately predicts y for previously unseen x

Can I eat this mushroom that **was not** in my training set?